

Synopsis

“Crime Prediction System”

Bachelor of Computer Applications



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CRIME PREDICTION SYSTEM

ABSTRACT:

The Crime Prediction System is an intelligent web-based application that uses Machine Learning techniques to analyze historical crime records of Karnataka and predict future crime occurrences. In traditional methods, police rely on manual reports and basic statistics without any predictive capability. This project uses a Random Forest Machine Learning algorithm to automatically predict the type of crime likely to occur in a specific district at a given hour.

The system displays interactive charts, year-wise crime reports (2015–2026), real NCRB case data, and a live intelligence dashboard. It provides district-wise crime analysis, police action plans, and a real-time risk indicator. The system has a secure web interface built using Python Flask where officers and admins can log in, predict crimes, and manage data.

Overall, this project helps law enforcement save time, plan patrol schedules better, and understand where and when crimes are most likely to occur. It makes crime analysis more accurate, automatic, and user-friendly.

INTRODUCTION:

A Crime Prediction System using Artificial Intelligence is an intelligent law enforcement support system that helps police automatically analyze, categorize, and predict crimes based on historical data.

It uses Machine Learning algorithms to identify crime patterns, predict future occurrences, and provide actionable intelligence to law enforcement agencies.

Instead of manually analyzing crime reports, the system processes historical crime data from 25 Karnataka districts and predicts the most likely crime type for any location and time combination.

By using machine learning techniques, the system can automatically classify, predict, and visualize crime data, giving police a clear understanding of crime patterns across Karnataka.

OBJECTIVES:

- Reduces crime rates through intelligent analysis.

WORKING / SYSTEM OVERVIEW:

Data Collection — Crime data is captured from CSV files containing district, hour, and crime type for 25 Karnataka districts

Data Processing — ML algorithms encode and classify crime data into categories like Murder, Theft, Kidnapping, Cyber Crime, etc.

Analysis — The system analyzes crime behavior by hour and location and predicts future occurrences

Visualization — Charts and graphs show crime distribution, hourly trends, and district rankings

Prediction — User inputs district, crime type, year, and time to get ML-based crime prediction with risk level

Live Intelligence — Real-time clock shows current hour risk level and most likely crime right now

TECHNOLOGIES USED:

Software Requirements:

- **Frontend:** HTML, CSS, JavaScript, Chart.js
- **Backend:** Python (Flask)
- **Database:** SQLite
- **AI/ML Tools:** scikit-learn (RandomForestClassifier), pandas, NumPy, LabelEncoder
- **Security:** Werkzeug (password hashing)
- **Tools:** VS Code

Hardware Requirements:

- **Processor:** Intel Core i3 or higher
- **RAM:** 4 GB or above
- **Storage:** 128 GB SSD or 256 GB SSD
- **Input Devices:** Keyboard, Mouse
- **Internet Connection:** Stable Wi-Fi (required for Chart.js CDN)

EXISTING SYSTEM:

The existing system of crime analysis mainly relies on manual work by police using basic statistical reports and spreadsheets. There is no predictive capability, which means police can only respond to crimes after they occur rather than preventing them in advance. Identifying crime patterns across multiple districts is time-consuming and error-prone. The system offers limited data visualization, making it difficult to understand crime trends. There is no integration with AI or Machine Learning to assist in decision-making, resource allocation, or proactive policing.

PROPOSED SYSTEM:

The proposed Crime Prediction System using AI aims to overcome the limitations of the existing system by introducing automation, intelligence, and visualization features. The proposed system combines Machine Learning (RandomForest), Flask backend, and a user-friendly dark-themed web interface to create an intelligent, automated, and interactive crime intelligence platform for Karnataka Police.

FEATURES:

- Real-time crime risk indicator with live clock on dashboard
- ML-based crime type prediction with risk level (Low/Medium/High)
- Year-wise crime reports (2015–2026) with district-wise breakdown
- Clickable crime cells showing real NCRB case data and police action plans
- Interactive charts — bar, pie, radar, and line charts
- District comparison and crime alert system
- Secure login with role-based access (Admin and Officer)
- Admin can upload new datasets and retrain the ML model automatically

MODULES:

User Login System: Each officer can have a secure account with hashed password authentication to access the crime intelligence dashboard.

ML-Based Crime Prediction: The system automatically predicts the most likely crime type for any district and hour using a trained RandomForest model. Provides estimated crime rate and risk level.

Predictive Analysis: The system analyzes year-wise crime trends (2015–2026) and shows projected crime reduction after system deployment to help police plan resources effectively.

Smart Alerts: Provides district-wise crime alerts showing top 5 high-risk districts requiring immediate police attention and resource deployment.

Data Visualization Dashboard: Displays charts and graphs showing crime type distribution, hourly crime patterns, district rankings, and year-wise crime trends.

Crime Detail Intelligence: Users can click any crime cell in the report to view real NCRB case data, injury information, peak hours, most affected districts, and a detailed police action plan.

ADVANTAGES:

- Saves time by automating crime pattern analysis
- Provides accurate crime predictions using ML
- Helps in patrol scheduling and resource allocation
- Easy to use and accessible from any browser
- Supports data-driven proactive policing
- Real NCRB case data makes it practically useful for officers

DISADVANTAGES:

- Depends on quality and completeness of crime data
- ML predictions may not always be 100% accurate
- Requires technical knowledge for setup and maintenance
- Data privacy concerns with sensitive crime information
- Model accuracy improves only with more training data

FUTURE SCOPE:

- Integration with real-time crime data from police databases
- Use of deep learning models (LSTM) for time-series crime prediction
- Mobile application development for field officers

- GPS-based crime tracking and live heatmap
- Integration with CCTV and IoT systems
- Natural Language Processing for FIR text analysis
- Prediction confidence score display using predict_proba

CONCLUSION:

The Crime Prediction System is an innovative solution designed to make law enforcement smarter, faster, and more proactive. This approach not only saves time but also helps police develop better patrol strategies, allocate resources efficiently, and prevent crimes before they happen.

With added features like live intelligence feed, real NCRB case data, and year-wise crime trend analysis, this project demonstrates how Artificial Intelligence can transform traditional policing into smart, data-driven crime prevention.

In conclusion, the proposed system provides a smart, accurate, and user-friendly way to predict and prevent crimes — a big step forward from traditional manual crime analysis methods.